

**RAMAIAH**

Institute of Technology

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DEPARTMENT OF MATHEMATICS

Sub Code:	AI 34/AD 34/CI 35/CY 35	Sub:	Discrete mathematical Structures	Test:	I
Time:	09.00 am -10.00 am.	Term:	20.11.23 to 02.03.24	Marks:	30
Date:	06.01.2024	Semester:	III	Sections:	AI/AD/CI/CY

Note: Answer any TWO full questions. Each main question carries 15 marks.

Q.No.	Questions	CO's	Blooms level	Marks
1.	(a) Construct the truth table for $\neg(p \vee (q \wedge r)) \Leftrightarrow (p \vee q) \wedge (p \rightarrow r)$.	CO1	L1	5
	(b) Prove by contrapositive "If $n = ab$ where a, b are positive integers then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$ ".	CO1	L3	5
	(c) Consider a set $A = \{1, 2, 3, 4\}$ and the relation $R = \{(1, 2)(2, 2)(2, 3)(1, 4)(4, 1)(4, 4)\}$ defined on A . Write the matrix of relation M_R and draw digraph of relation. Also find in-degree and out-degree of all vertices.	CO2	L3	5
2.	(a) Define anti-symmetric relation. Let Z be the set of all integers and $R = \{(a, b)/a < b\}$. Discuss whether R is reflexive, anti-symmetric and transitive.	CO2	L2	5
	(b) Prove by Mathematical induction " $11^n - 4^n$ is divisible by 7, $\forall n \in \mathbb{Z}^+$ ".	CO1	L4	5
	(c) Write the converse, inverse and contrapositive of "If n is not a prime number then it is a composite number".	CO1	L3	5
3.	(a) Define Poset. Let $A = \{1, 2, 3, 4, 6, 12\}$ and on A define the relation R by aRb if a divides b . Verify (A, R) is a Poset or not?	CO2	L4	5
	(b) Define equivalence relation. The relation R defined on the set of all integers Z as $R = \{(x, y)/x, y \in \mathbb{Z}, \text{ and } (x - y) \text{ is an even integer}\}$. Verify that R is an equivalence relation or not?	CO2	L2	5
	(c) Without constructing the truth table prove $[\neg p \wedge (p \vee q)] \rightarrow q$ is a Tautology.	CO1	L3	5